



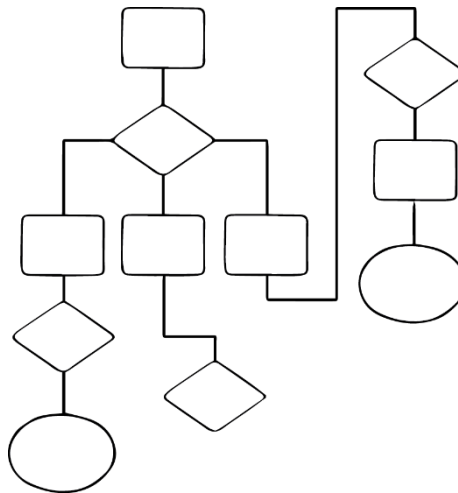
Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

**CONTENT**

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**Let's Start!****Class Diagrams and Their Features****Learning Outcome:**

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.  
Give detailed information related to UML and common related topics.

**Can Do Statements:**

- I can explain how the class diagram works.
- I can give details about class diagram.
- I can identify key features related to class diagram.

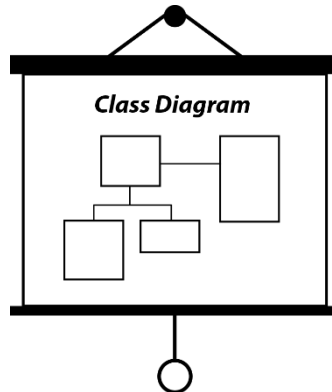
**Ask your partner:**

- Why do you think it is important to organize your ideas hierarchically?
- What is the easiest way to organize a diagram? Explain.



## Let's Explore!

Two students are talking about class diagrams.



*Instructions: Read the text below.*

- Tanner:** So, Justin can we please review the information we have about class diagrams?
- Justin:** Of course, Tanner! We should do that, because we have the longest exam next week!
- Tanner:** Okay, I can see in my notes that class diagrams can show a static view of the system by displaying the classes, particular attributes, behaviors, operations, and interactions between objects. It's one of the most popular modeling standards in UML. In this kind of models, a class refers to the attributes and behaviors that objects can generate. It also contains possible messages, constraints, stereotypes or tagged values within the object.
- Justin:** I see. We should try to identify their key features; that is the fastest way to understand them. There's plenty of information online!
- Tanner:** Definitely. It's important to note that class diagrams can show a static structure of the software, it doesn't take time into account. The clearest way to make use of these diagrams is to present system types, instances, and the way they should relate and interact. We can even see the internal structure of those classes in a hierarchy.
- Justin:** That's one of the easiest definitions of class diagrams! Another characteristic is that we can display classes in boxes that have three sections; the first one has the name, the second one includes the attributes of the class and the last one has the operations that the class can execute.
- Tanner:** One of the attributes of these diagrams is that, by showing relationships, there are conditions or operations that we need to complete for objects to interact.
- Justin:** I also see that the strongest advantage of these diagrams is when designers need to document the system's code, because they present a solid foundation for defining tasks and workflow.
- Tanner:** Should we keep looking for more information?
- Justin:** No, I think we're ready for that exam now!

**Let's Practice!**

What can you tell me about class diagrams?

*Instructions: Read the sentences. Write the missing word(s) in the blank. Use: should try, can show, can generate, should draft.*

1. Class diagrams \_\_\_\_\_ a static view of the software model.
2. A class refers to the attributes and behaviors that objects \_\_\_\_\_.
3. We \_\_\_\_\_ to identify the key features of a class in UML language.
4. We \_\_\_\_\_ what the software objects do by means of a class diagram.

**Let's Practice!**

The Best Reasons to Use UML Class Diagrams

*Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. There is only one possible answer. Use: strongest advantages / clearest / most popular / best way*

Among the many modeling standards that are out there when talking about object-oriented programming, one of the \_\_\_\_\_ ones is the UML class diagrams. One of the \_\_\_\_\_ of using class diagrams is the ease of graphing the relation between objects and classes within a system; Some people think they are the \_\_\_\_\_ to perform such task because they allow the designer to correlate specific classes to their attributes and methods. . Another great advantage is that it is the \_\_\_\_\_ way to present systems, types and their relation.





## Extension Activity

It time to put into practice what you've learnt about UML class diagrams.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Look for information about the different kinds of class diagrams
- Draw a diagram where you show what you can do with the different kinds of class diagrams and when you should use them.
- Share your diagram with your classmates.

### Self-Assessment

Yes

No

- |                                                                                                 |                       |                       |
|-------------------------------------------------------------------------------------------------|-----------------------|-----------------------|
| 1. I research on class diagrams.                                                                | <input type="radio"/> | <input type="radio"/> |
| 2. I create a diagram where I explain what I can do with the different kinds of class diagrams. | <input type="radio"/> | <input type="radio"/> |
| 3. I share my findings with my classmates.                                                      | <input type="radio"/> | <input type="radio"/> |



## Target Words

- **static (adjective):** something that cannot move or remains in the same place  
e.g. Okay, I can see in my notes that class diagrams can often show a static view of the software model
- **within (adverb):** inside, in the interior of  
e.g. It also contains possible messages, constraints, stereotypes or tagged values within the object.
- **tagged value (noun):** in programming, it refers to an additional value that goes beyond the support of UML  
e.g. It also contains possible messages, constraints, stereotypes or tagged values within the object.
- **key feature (noun):** one of the most important characteristics or qualities of something  
e.g. We should try to identify their key features; that is the fastest way to understand them.
- **take into account (expression):** to consider, to contemplate or to envisage  
e.g. It's important to note that class diagrams can show a static structure of the software, it doesn't take time into account.
- **workflow (noun):** the way work should be executed in time or order  
e.g. They present a solid foundation for defining tasks and workflow.



## Learning Points

### Modal Verbs (Review)

Modal verbs allow you to express ability, obligation, desire, and permission to do something when you use them with other verbs.

#### Can

You use this modal verb to talk about possibilities, abilities and requests (informal). In order to develop ideas including the modal verb "**can**," you need to use the following structure:

#### **Subject + Modal verb(can) + main verb in base form + complement**

In negative sentences, you place the "not" adverb after the modal verb.

Examples:

Class diagrams **can** display interactions between objects.  
Structure diagrams **cannot/can't** be used for this project.

#### Should

This modal verb is used to give suggestions and advice. In order to develop ideas including the modal verb "**should**," you need to use the following structure:

#### **Subject + Modal verb (should) + main verb in base form + complement**

In negative sentences, you use **Not** after *Should*.

Examples:

We **should** use a more detailed behavior diagram.  
They **shouldn't/should not** rely so much in drafts for modeling software.

### Examples

- *Of course, Tanner! We should do that, because we have the longest exam next week!*
- *Okay, I can see in my notes that class diagrams can show a static view of the system by displaying the classes, particular attributes, behaviors, operations, and interactions between objects.*





## Learning Points

### Superlatives

You can use **Superlatives** to talk about a noun that the highest degree of a quality.

In order to use superlatives with **short** adjectives, you need to use the definite article **The**, and the adjective with the **-est** suffix. Remember, short adjectives have only one or two syllables.

Examples:

**The fastest** way to develop software is by using the appropriate modeling tool. (fast)  
Digital tools offer **the simplest** approach to software modeling. (simple)

**Note:** There are a few exceptions to this rule. With irregular adjectives, the superlative form changes. Here are a few examples:

| Adjective | Superlative       |
|-----------|-------------------|
| good      | best              |
| bad       | worst             |
| far       | furthest/farthest |

Example:

Class diagrams are the **best** diagrams we can use for this purpose.

When you have long adjectives (two or more syllables), you need to use the definite article **The**, the particle **most**, and the adjective.

Examples:

**The most challenging** part of using a class diagram is to follow it through. (challenging)

Even **the most advanced** diagram can't predict how the final software will work.(advanced)

### Examples

- *That's one of the easiest definitions of class diagrams!*
- *I also see that the strongest advantage of these diagrams is when designers need to document the system's code, because they present a solid foundation for defining tasks and workflow.*





## Learning Points

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### What is a Diagram?

A diagram is a graphic representation of something. Sometimes, you can also refer to them using the word Graph.

In software design, planning and development, diagrams help engineers and designers to have a draft of the set of instructions, commands and actions that a piece of software can have.

UML is a standardized way of making software development diagrams. There are many of them, each one serving a different purpose depending on the designer's needs. However, diagrams in UML are divided in two major groups, which are Structural, Behavior and Interaction.

Among structural ones, you can find Class, deployment, object, package and profile diagrams. This kind presents the information that must be included in the system as well as the structure it must have.

Behavior diagrams show the actions that need to occur within the system, as well as its functionality. Interaction diagrams are considered to be a part of behavioral ones, and they focus more on control and dataflow.